

SQL Subquery

Unit-3

subquery

- A subquery in SQL is a query nested inside another SQL query. It allows complex filtering, aggregation and data manipulation by using the result of one query inside another. They are an essential tool when we need to perform operations like:
- Filter rows based on results from another query.
- Apply aggregate functions like SUM, COUNT, or AVG dynamically.
- Update data using values from other tables.
- Delete rows based on conditions returned by another query.

Example

StudentID	Name	Score
1	Alice	88
2	Daniel	92
3	Maria	79

- `SELECT * FROM Students`
- `WHERE Score > ( SELECT AVG(Score) FROM Students );`

StudentID	Name	Score
1	Alice	88
2	Daniel	92

- The subquery calculates the average score of all students.
- The main query returns only the students whose score is higher than that average.

Syntax:

- SELECT column_name
- FROM table_name
- WHERE column_name expression operator
- (SELECT column_name FROM table_name WHERE ...);

SQL Clauses for Subqueries

- Clauses that can be used with subqueries are:
- Filters rows in the outer query based on the results returned by the WHERE clause.
- Treats the subquery as a temporary (derived) table that can be queried like a normal table using the **FROM** clause.
- Filters grouped or aggregated results using values produced by the HAVING clause.

Types of Subqueries

EmployeeID	Name	Salary	DepartmentID
1	Alice	80000	10
2	Bob	60000	10
3	Charlie	90000	20
4	Diana	75000	20
5	Evan	50000	30

DepartmentID	DepartmentName	Location
10	HR	New York
20	Engineering	Chicago
30	Sales	New York

Single-Row Subquery (Scalar Subquery)

- A single-row subquery is a subquery that returns only one value.
- Returns exactly one row as the result.
- Commonly used with comparison operators such as =, >, <

Example:

- `SELECT * FROM Employees`
- `WHERE Salary = (SELECT MAX(Salary) FROM Employees);`

EmployeeID	Name	Salary	DepartmentID
3	Charlie	90000	20

- Finds the highest salary in the Employees table.
- Returns the employee(s) whose salary matches that maximum value.

Multi-Row Subquery

- A multi-row subquery is a subquery that returns more than one value.
- Returns multiple rows as the result.
- Requires operators that can handle multiple values, such as IN, ANY or ALL

Example:

- `SELECT * FROM Employees`
- `WHERE DepartmentID IN (SELECT DepartmentID FROM Departments WHERE Location = 'New York');`

EmployeeID	Name	Salary	DepartmentID
1	Alice	80000	10
2	Bob	60000	10
5	Evan	50000	30

- Identifies all departments located in New York.
- Returns the employees who belong to any of those departments.

Correlated Subquery

- A correlated subquery is a subquery that depends on the outer query for its values.
- A dependent subquery: it references columns from the outer query.
- Executed once for each row of the outer query, making it slower for large datasets.

Example:

- SELECT e.Name, e.Salary
- FROM Employees e
- WHERE e.Salary > (SELECT AVG(Salary)
- FROM Employees
- WHERE DepartmentID = e.DepartmentID);

Name	Salary
Alice	80000
Charlie	90000

- Calculates the average salary within each employee's department.
- Returns employees whose salary is higher than their department's average.

Examples of Using SQL Subqueries

Student_info Table

NAME	ROLL_NO	LOCATION	PHONE_NUMBER
James	101	New York	9876543210
Sophia	102	London	8765432109
Daniel	103	Toronto	7654321098
Emma	104	Sydney	6543210987
Oliver	105	Berlin	5432109876

Student_section Table

NAME	ROLL_NO	SECTION
Emma	104	A
Oliver	105	B
Sophia	102	A

Example 1: Fetching Data Using Subquery in WHERE Clause

- SELECT NAME, LOCATION, PHONE_NUMBER
- FROM Student_Info
- WHERE ROLL_NO IN (
 - SELECT ROLL_NO
 - FROM Student_Section
 - WHERE SECTION = 'A'
-);

NAME	LOCATION	PHONE_NUMBER
Sophia	London	8765432109
Emma	Sydney	6543210987

- The subquery NAME, LOCATION and PHONE_NUMBER, WHERE SECTION = 'A' finds roll numbers of students in section A.
- The outer query then uses these roll numbers to fetch details from the Student table.
- Thus, only Sophia and Emma are returned, since they are in section A.

Example 2: Using Subquery with DELETE

- DELETE FROM Student_Info
- WHERE ROLL_NO IN (
 - SELECT ROLL_NO FROM Student_Info WHERE ROLL_NO <= 101 OR ROLL_NO = 201);

NAME	ROLL_NO	LOCATION	PHONE_NUMBER
Sophia	102	London	8765432109
Daniel	103	Toronto	7654321098
Emma	104	Sydney	6543210987
Oliver	105	Berlin	5432109876

- The subquery selects roll numbers 101 and 201.
- The outer query deletes students having those roll numbers.

Example 3: Using Subquery with UPDATE

- UPDATE Student_Info
- SET NAME = 'Geeks'
- WHERE LOCATION IN (
 - SELECT LOCATION
 - FROM Student_Info
 - WHERE LOCATION IN ('London', 'Berlin')
-);

NAME	ROLL_NO	LOCATION	PHONE_NUMBER
Geeks	102	London	8765432109
Daniel	103	Toronto	7654321098
Emma	104	Sydney	6543210987
Geeks	105	Berlin	5432109876

- The subquery selects locations 'London' and 'Berlin'.
- The outer query updates the NAME field for students whose location matches those values.

- SELECT NAME, PHONE_NUMBER
- FROM (
 - SELECT NAME, PHONE_NUMBER, LOCATION
 - FROM Student_Info
 - WHERE LOCATION LIKE 'T%'
-) AS subquery_table;

NAME	PHONE_NUMBER
Daniel	7654321098

- The subquery (SELECT NAME, PHONE_NUMBER, LOCATION FROM Student WHERE LOCATION LIKE 'T%') fetches students whose location starts with "T" (Toronto).
- The outer query then selects only NAME and PHONE_NUMBER from this derived table.

Why the table name is not written in the outer query?

- The **outer query** is not directly accessing a table.
It is accessing the **result of the inner query (subquery)**.
- The inner query creates a **temporary result table (derived table)**
- So the outer query actually reads from this **temporary table**.

Example 5: Subquery with JOIN

- SELECT s.NAME, s.LOCATION, ns.SECTION
- FROM Student_Info s
- INNER JOIN (
 - SELECT ROLL_NO, SECTION
 - FROM Student_Section
 - WHERE SECTION = 'A') ns
- ON s.ROLL_NO = ns.ROLL_NO;

NAME	LOCATION	SECTION
Emma	Sydney	A
Geeks	London	A

- The subquery extracts roll numbers of students in section A.
- Joining this with the Student_Info table on ROLL_NO returns Emma and Sophia along with their locations and section.

